Consider the four letters A, B, C, D and the number of row arrangement of these 4 letters can be done in 4! Ways. Of those 4! arrangements, the arrangements ABCD, BCDA, CDAB, DABC are one and the same when they are arranged along a circle.

So, the number of permutation of '4' things along a circle is $\frac{4!}{4} = 3!$

In general, circular permutation of n different things taken all at the time = $(n-1)!$

NOTE

If clock wise and anti-clockwise circular permutations are considered to be same (identical), the number of circular permutation of n objects taken all at a time is $\frac{(n-1)!}{2}$

Example 2.22

In how many ways 8 students can be arranged

(i) in a line (ii) along a circle

Solution

(i) Number of permutations of 8 students along a line = $8P_8 = 8!$

(ii) When the students are arranged along a circle, then the number of permutation is $(8-1)! = 7!$

Example 2.23

In how many ways 10 identical keys can be arranged in a ring?

Solution

Since keys are identical, both clock wise and anti-clockwise circular permutation are same.

The number of permutation is

$$\frac{(n-1)!}{2} = \frac{(10-1)!}{2} = \frac{9!}{2}$$

Example 2.24

Find the rank of the word 'RANK' in dictionary.

Solutions

Here maximum number of the word in dictionary formed by using the letters of the word 'RANK' is 4!

The letters of the word RANK in alphabetical order are A, K, N, R

DO YOU KNOW?

Rank of word in dictionary

*The **rank** of a word in a dictionary is to arrange the words in alphabetical order. In this dictionary the word may or may not be meaningful.*

Number of words starting with A =3! =6

Number of words starting with K =3! =6

Number of words starting with N =3! =6

Number of words starting with RAK =1! = 1

Number of words starting with RANK = 0! = 1

∴ Rank of the word RANK is

6 + 6 + 6 + 1+1= 20

Exercise 2.3

1. If $nP_4 = 12(nP_2)$, find n.
2. In how many ways 5 boys and 3 girls can be seated in a row, so that no two girls are together?
3. How many 6-digit telephone numbers can be constructed with the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 if each numbers

02_11th_BM-STAT_Ch-2-EM.indd 35 21-04-2020 12:18:24 PM